

SESSION 20

Capstone: Doctor Finder Project (Integration & Cleanup)

Topics Covered: Includes Registration, Login, OTP, Forgot Password, Profile Management, Search Doctors, Filters, Appointment booking, Google Maps, Payment gateway, Deployment

Activities: Final code on GitHub, Final deployment URL, Project documentation

Task 1: Add feature to project allowing users to search restaurants by cuisine and location (Zomato search).

Explanation & Approach: *Combines multiple query parameters with Django ORM field lookups.*

```
# views.py
from django.shortcuts import render
from .models import Restaurant

def search_restaurants(request):
    cuisine = request.GET.get('cuisine', '')
    location = request.GET.get('location', '')

    results = Restaurant.objects.all()
    if cuisine:
        results = results.filter(cuisine__name__icontains=cuisine)
    if location:
        results = results.filter(location__icontains=location)

    return render(request, 'capstone/search_results.html', {'results': results})
```

Task 2: Implement login and OTP verification system for movie ticket booking app expiring after 3 minutes.

Explanation & Approach: *Combines authentication and temporary session OTP expiration.*

```
# OTP Verification with 3-min timeout
def verify_login_otp(request):
    user_otp = request.POST.get('otp')
    saved_otp = request.session.get('login_otp')

    if saved_otp and user_otp == saved_otp:
        del request.session['login_otp']
        return redirect('dashboard')
    else:
        return render(request, 'accounts/otp.html', {'error': 'OTP Expired or Invalid!'})
```

Task 3: Build user profile management page for Flipkart-style app allowing users to update name, email, delivery address, profile pic.

Explanation & Approach: *Full user profile management combining auth User and Profile models.*

```

@login_required
def update_flipkart_profile(request):
    if request.method == 'POST':
        request.user.first_name = request.POST['name']
        request.user.email = request.POST['email']
        request.user.save()

        profile = request.user.profile
        profile.address = request.POST['address']
        if 'profile_pic' in request.FILES:
            profile.profile_pic = request.FILES['profile_pic']
        profile.save()
        return redirect('profile_view')

```

Task 4: Integrate Google Maps into cab booking page — show current location on map and select pickup location by clicking on map.

Explanation & Approach: *Google Maps JavaScript API click event handler extracting clicked lat/lng coordinates.*

```

<!-- Google Maps Click Selection -->
<div id="map" style="height: 400px; width: 100%;"></div>
<input type="text" id="pickup-lat" name="lat" readonly>
<input type="text" id="pickup-lng" name="lng" readonly>

<script>
function initMap() {
    var defaultLoc = {lat: 12.9716, lng: 77.5946};
    var map = new google.maps.Map(document.getElementById('map'), {
        zoom: 13,
        center: defaultLoc
    });

    var marker = new google.maps.Marker({
        position: defaultLoc,
        map: map
    });

    map.addListener('click', function(e) {
        var clickedLoc = e.latLng;
        marker.setPosition(clickedLoc);
        document.getElementById('pickup-lat').value = clickedLoc.lat();
        document.getElementById('pickup-lng').value = clickedLoc.lng();
    });
}
</script>
<script src="https://maps.googleapis.com/maps/api/js?key=YOUR_KEY&callback=initMap" async
defer></script>

```

Task 5: Deploy completed project to PythonAnywhere, push final code to GitHub repository, share both live URL and GitHub link.

Explanation & Approach: *Final capstone summary containing public repository link and deployed web URL.*

```
# Capstone Project Final Deployment Summary
```

- GitHub Repository URL: https://github.com/yourusername/DoctorFinder_Capstone
- Live Web Deployment URL: <https://yourusername.pythonanywhere.com/>

Project Status: Fully Tested, Verified, Deployed & Live!